

Mass Weights of Standard Model Particles

Topological Mass Suppression as an Internal Weighting — Scope and Status

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Abstract

The rope framework attributes particle mass hierarchies to a topological/internal weighting rather than to Yukawa couplings. This paper states what the model computes in this direction and what remains conjectural. The mass-suppression mechanism is internal to the knot structure, and the charged-lepton sector is captured by the Koide-phase relation reported separately (with its documented conditional-success caveat). Other Standard-Model mass weights are, at present, modeled or open rather than derived. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants; the lepton-ratio caveat below is pinned in the test suite.

Scope of this paper

CLAIM: particle mass in the rope model is an internal/topological weighting; the charged-lepton case is captured by the Koide relation (conditionally).

NOT CLAIMED: a complete, first-principles mass spectrum for all Standard-Model particles. Most weights are Open.

1. Mass as Internal Weighting

In the rope picture, the large hierarchy of particle masses reflects how strongly a given knot configuration is internally suppressed — a topological weighting — rather than a set of independent coupling constants. This reframes the hierarchy problem as a question about knot structure.

2. What Is and Is Not Established

- Charged leptons: captured by the Koide-phase relation, conditionally on the measured Weinberg angle (see the Lepton Masses paper; the failure with the model's own θ_W is pinned).
- Quarks and bosons: the internal-weighting picture is stated but the specific mass weights are not derived here; they are open.

Status

Mass = internal topological weighting (framing) — Modeled.

Charged-lepton weights — Conjecture (conditional, see Lepton Masses).
Full SM mass spectrum — Open.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.